

VX-850

SEVERE-SERVICE VALVE / Flow Control

A severe-service control reference for high-energy, high-temperature, or erosive duties where trim selection, pressure drop, materials, and inspection access must be resolved early.



PLATFORM ROLE	REFERENCE STATUS	DOCUMENT USE
High-energy and erosive service valve platform	Illustrative project concept	Early engineering conversation

TECHNICAL SPECIFICATION

PARAMETER	REFERENCE ENVELOPE
SIZE RANGE	DN50-DN500
PRESSURE CLASS	ASME 150-1500
TEMPERATURE	-50 C to 550 C
TRIM	Tungsten carbide / Ceramic
BODY	Forged alloy steel

The values above are a reference envelope for concept discussions. Confirm duty, materials, performance, and compliance requirements against the approved project specification before procurement or manufacture.

VX-850 / ENGINEERING NOTES

SEVERE-SERVICE VALVE / CONFIGURATION FRAMEWORK

The platform is configured around the operating envelope rather than selected from a single catalogue point. Use the prompts below to structure the application review.

DESIGN INTENT AND INTERFACES

ENGINEERING LENS	WHAT TO RESOLVE
Control intent	Manage a high-energy pressure drop while keeping the selected trim and body materials aligned to the service conditions.
Process attention	Pressure drop, flashing or cavitation risk, erosion mechanism, thermal gradients, actuator sizing, and noise review.
Release boundary	Trim geometry and materials are selected from the approved process data; this reference does not certify a service limit.

CONFIGURATION REVIEW

ENGINEERING LENS	WHAT TO RESOLVE
Trim path	Document the pressure-drop case, phase behavior, particle loading, and expected wear mechanism before final trim selection.
Thermal path	Review body, bonnet, packing, actuator, and instrument temperature exposure together, including warm-up and shutdown states.
Maintenance	Provide inspection access and a replacement strategy for the trim elements that are expected to see the highest energy.

ENGINEERING HANDOFF CHECKLIST

01	Pressure-drop and phase cases documented
02	Erosion, cavitation, and noise review completed
03	Trim, body, packing, and actuator basis agreed
04	Inspection and replacement access included in layout

REFERENCE NOTE	This document is a polished portfolio reference, not a certified data sheet. Exact performance, materials, interfaces, testing, and applicable requirements remain project-specific and require written confirmation.
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